PAVAN SRINIVAS DOMA

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SKILLS

LLMs/ Gen AI: RAG Systems, LangGraph, Prompt Engineering, Fine-tuning, LLM Evaluation, Hugging Face, Vector Databases
Data and Frameworks: PyTorch, TensorFlow, Keras, Scikit-Learn, Pandas, NumPy, Matplotlib
Languages and Tools: Python, SQL, AWS (EC2, ECS, Lambda), Git, FastAPI,
Technical Skills: Machine Learning, Natural Language processing, Data Analytics
Cloud & Infrastructure: AWS ECS, Docker, containerization, distributed systems, production deployment, reliability optimization

EDUCATION

University at Buffalo, SUNY — **Buffalo, USA**
Master's in Engineering science — Jan 2026

S.R.M Institute of Science and Technology, SRM University — **Chennai, India**
Bachelor of Technology in Artificial Intelligence. — May 2024

EXPERIENCE

DATA ENGINEER - OPERATIONS LEAD — **Buffalo, USA**
University at Buffalo — Sep 2024 - Dec 2025

- Collaborated with operations team to design & maintain normalized database schema (people, inventory, transactions), implementing validation rules and data quality checks that ensured 100% transactional integrity-foundation for reliable reporting and decision-making across the organization.
- Reduced query runtime by 40% by developing real-time KPI dashboards using advanced SQL (window functions, CTEs, aggregations) and optimizing queries.
- Engineered production resource allocation pipeline leveraging SQL window functions and time-series decomposition, collaborating with stakeholders to validate business logic demonstrated end-to-end ownership of data flow from extraction to actionable insights for leadership.
- Optimized ETL pipeline performance by real-time operational decision-making that improved asset utilization efficiency by 25%, through SQL query refactoring and schema redesign enabling directly supporting business scalability goals.
- Standardized & documented ETL pipelines for stock tracking/allocation, establishing data quality gates and error-handling protocols reduced data ingestion errors by 30%, shortened refresh cycles, and created runbooks enabling team to maintain system reliability and onboard new engineers faster.

ACADEMIC PROJECTS

RETRIEVAL-AUGMENTED CONVERSATIONAL AI SYSTEMS (LLMS)

- Designed production-scale RAG system, working with ML engineers to validate semantic relevance (97.5% accuracy) containerized deployment on AWS ECS with monitoring infrastructure, demonstrating end-to-end ownership of data pipelines supporting enterprise AI workflows.
- Implemented context engineering, memory management, and structured output validation to ensure reliable, context-aware responses across enterprise workflows.

CREDIT SCORING & CUSTOMER SEGMENTATION (DATA-DRIVEN SYSTEMS)

- Developed a credit risk scoring system collaborating cross-functionally to define feature engineering strategy
- Validated business logic demonstrated understanding of data quality trade-offs and working with senior engineers to compare 5 model approaches via MLflow experiment tracking.

CONSUMER COMPLAINT CLASSIFICATION (SCALABLE NLP SYSTEM)

- Launched an intent classifier for banking customer queries across 77 intent classes, achieving 89.5% accuracy.
- Implemented a confidence-threshold pipeline allowing 49.6% of predictions to skip manual review while maintaining 99.8% accuracy on auto-approved cases.

CERTIFICATIONS

- Claude with Amazon Bedrock (Generative AI & LLM Application Development) — July 2026
- AWS Cloud Practitioner (Cloud platform deployment & AI services) — Jan 2026